

- 5 (a)** *either* $\phi = BA \sin \theta$ M1
 where A is the area (through which flux passes)
 θ is the angle between B and (plane of) A A1
or
 $\phi = BA$ (M1)
 where A is area normal to B (A1) [2]
- (b)** graph: V_H constant and non zero between the poles and zero outside M1
 sharp increase/decrease at ends of magnet A1 [2]

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(c) (i) (induced) e.m.f. proportional to rate of change of (magnetic) flux (linkage) M1
A1 [2]

(ii) short pulse on entering and on leaving region between poles M1
pulses approximately the same shape but opposite polarities A1
e.m.f. zero between poles and outside A1 [3]

6 (c) (i) connection to 'top' of resistor labelled as positive B1 [1]